

DAV-R WS2024 GRADUATION PROJECT
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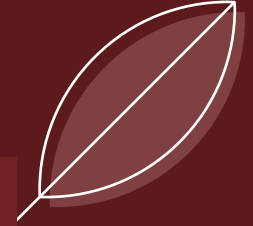
*Wrangling of Antibiotic Resistance Microbiology Dataset for Analysis of
Klebsiella Pneumoniae Trends*

ABSTRACT



- Antimicrobial resistance continues to increase while the pipeline for new antibiotic development is drying up; after only eight decades of antibiotic use, bacterial infections that once were easily treated are becoming untreatable. Antimicrobials have enabled the advancement of many areas of medical practice. The successful outcomes of many surgical procedures and immunosuppressive treatments depend on antibiotic prophylaxis and the ability to treat infectious complications. Antibiotic resistance correlates with antibiotic use, so improved antimicrobial stewardship, with better prevention and diagnosis of infection, can help to conserve the currently available antimicrobial agents. Significant global action and investment, from both public and private sector funding, is required if the development of new anti-infectives is to keep pace with increasing resistance.

PIPELINE STRUCTURE



1. Data source and Acquisition.
2. Data Wrangling (cleaning & Preprocessing)
3. Exploratory Data Analysis (EDA)
4. Statistical Analysis
5. Visualization & Reporting

DATA SOURCE AND ACQUISITION

The Antibiotic Resistance Microbiology Dataset (ARMD) is a de-identified resource derived from electronic health records (EHR) that facilitates research into antimicrobial resistance (AMR). It provides a comprehensive overview of microbiological cultures, including urine, respiratory, and blood cultures. This dataset includes 283,715 unique adult patients and features detailed information on culture results, identified organisms, antibiotic susceptibility, and associated demographics and clinical data. The dataset was meticulously constructed through a multi-step process designed to enhance data quality and relevance. Key attributes include organism identification, susceptibility patterns for 55 antibiotics, implied susceptibility rules, and de-identified patient information. This dataset supports studies on antimicrobial stewardship, causal inference, and clinical decision-making. The dataset integrates microbiology laboratory results, demographic data, clinical encounters, antibiotic exposures, and socioeconomic indicators to comprehensively analyze AMR patterns.

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DATA WRANGLING (CLEANING & PREPROCESSING)

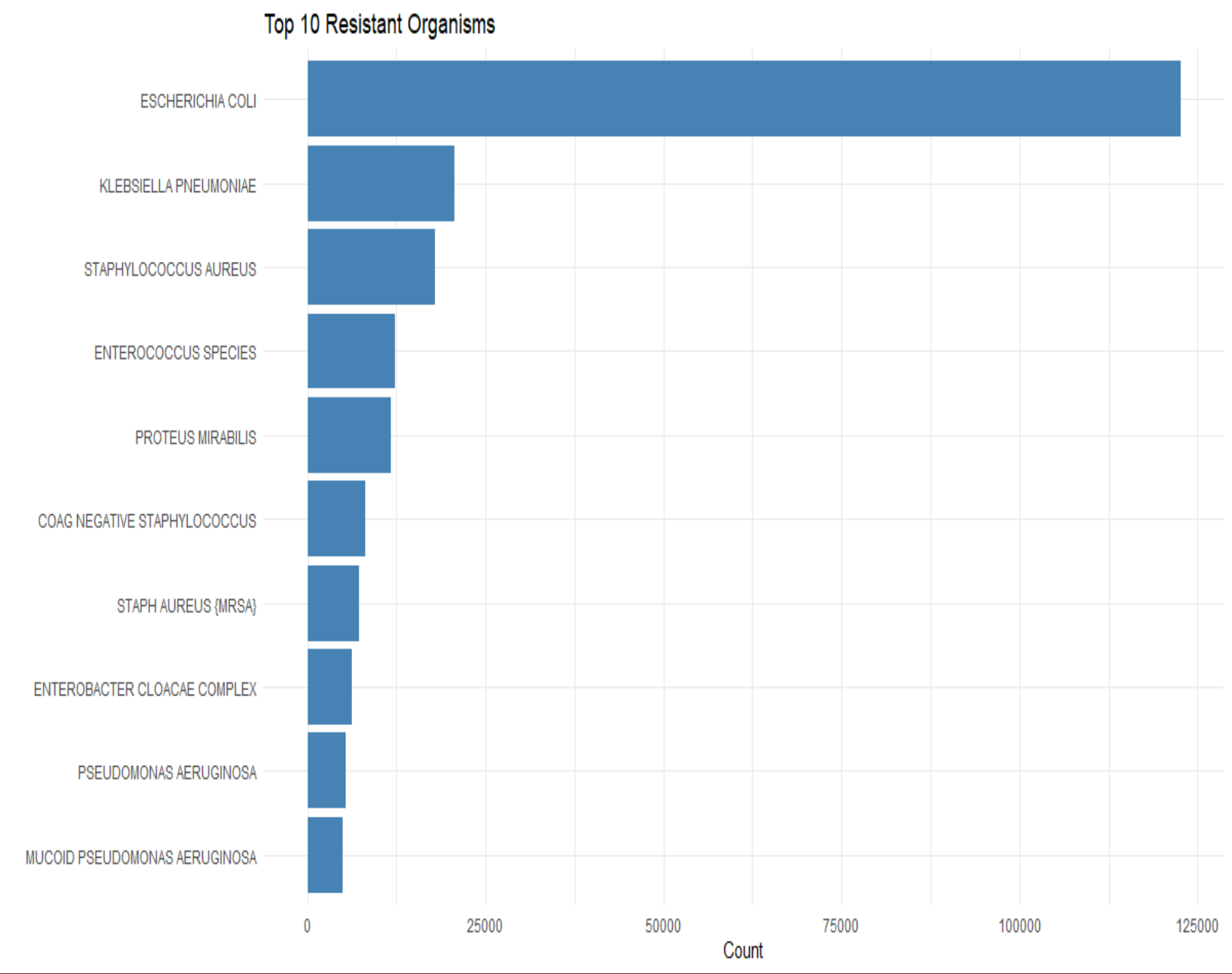
1. *Initial exploration for missing values and data summarization.*
2. *Handling missing data:* Empty cells in the dataset are represented as "null" to ensure clarity.
3. *Standardization.*
4. *Filtering & Subsetting: selecting data of interest and merging between datasets using a common key column.*



EXPLORATORY DATA ANALYSIS

DISTRIBUTION SHOWS THE TOP TEN MOST FREQUENTLY ISOLATED MICROORGANISMS

As shown, Klebsiella pneumoniae is the second most resistant microorganism to antibiotics. The danger comes from it being the second most common cause (behind Escherichia coli) of community- and hospital-acquired Gram-negative bloodstream infection, which is called nosocomial infection. K. pneumoniae bloodstream infections usually arise as a complication of focal urinary, gastrointestinal, or respiratory tract infections. However, occasionally they can arise without a definable source.



HERE'S A PICTURE FROM THE NEW YORK ENDOVASCULAR SUMMIT, BACK IN DECEMBER 2024 LECTURE: INTERPRETATION OF A C&S REPORT BY DR. MARK A KOSINSKI

A culture and sensitivity report from a Biochemical lab in Egypt, back to 2017.

The isolated organism is Klebsiella Pneumoniae, shown here resistant to all kinds of antibiotics, except colistin, which is nephrotoxic.

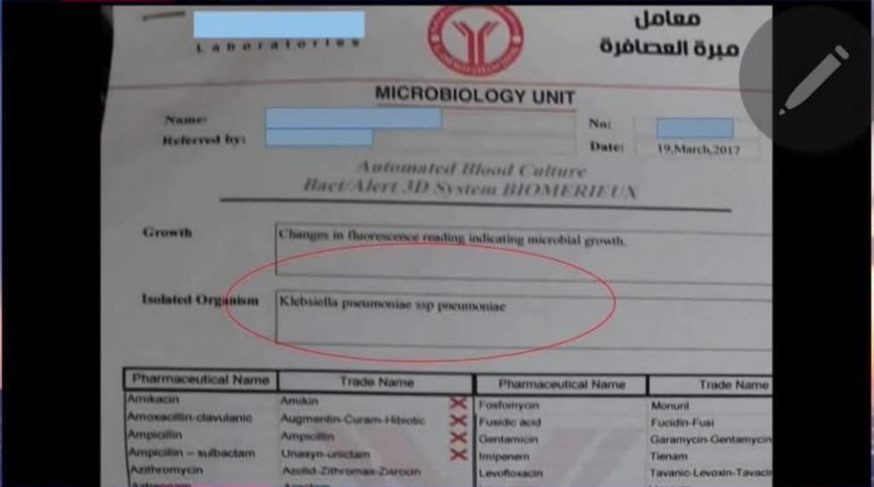


Amr El-Mahallawy
3h · 🌐

دي محاضرة في مؤتمر New York endovascular summit

المحاضر بيتكلم عن ال antibiotic resistance

راح جاب صورة تحليل من مبرة العصفرة. ...(((

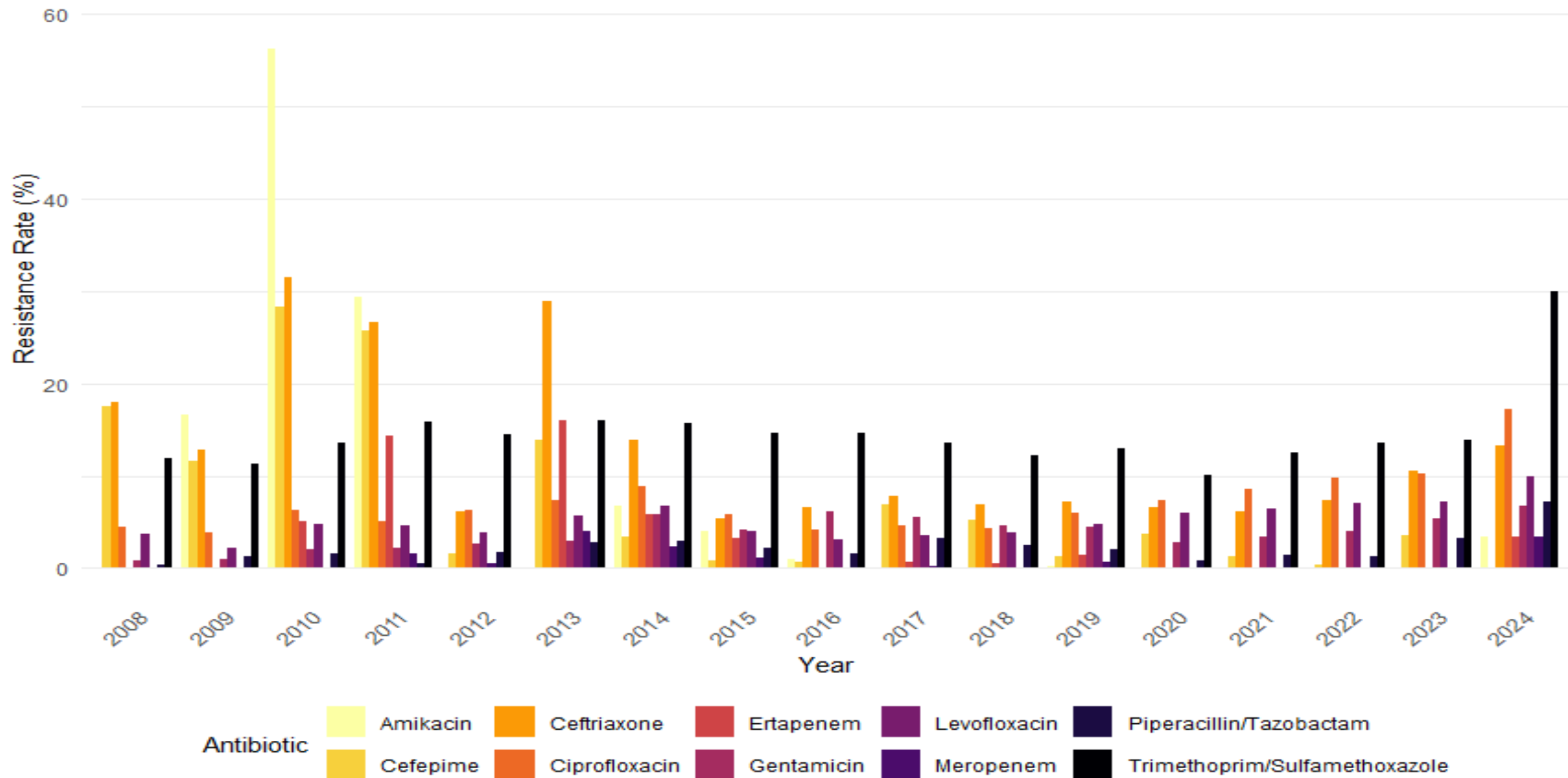


Pharmaceutical Name	Trade Name	Pharmaceutical Name	Trade Name
Amikacin	Amikin	Fosfomycin	Monuril
Amoxicillin-clavulanic	Augmentin-Curam-Hibiotic	Fusidic acid	Fucidin-Fusi
Ampicillin	Ampicillin	Gentamicin	Garamycin-Gentamycin
Ampicillin – sulbactam	Unasyn-uristam	Imipenem	Tienam
Azithromycin	Azolid-Zithromax-Zaroon	Levofloxacin	Tavanic-Levoxin-Tavacin
Autopocaine	Autopocaine		

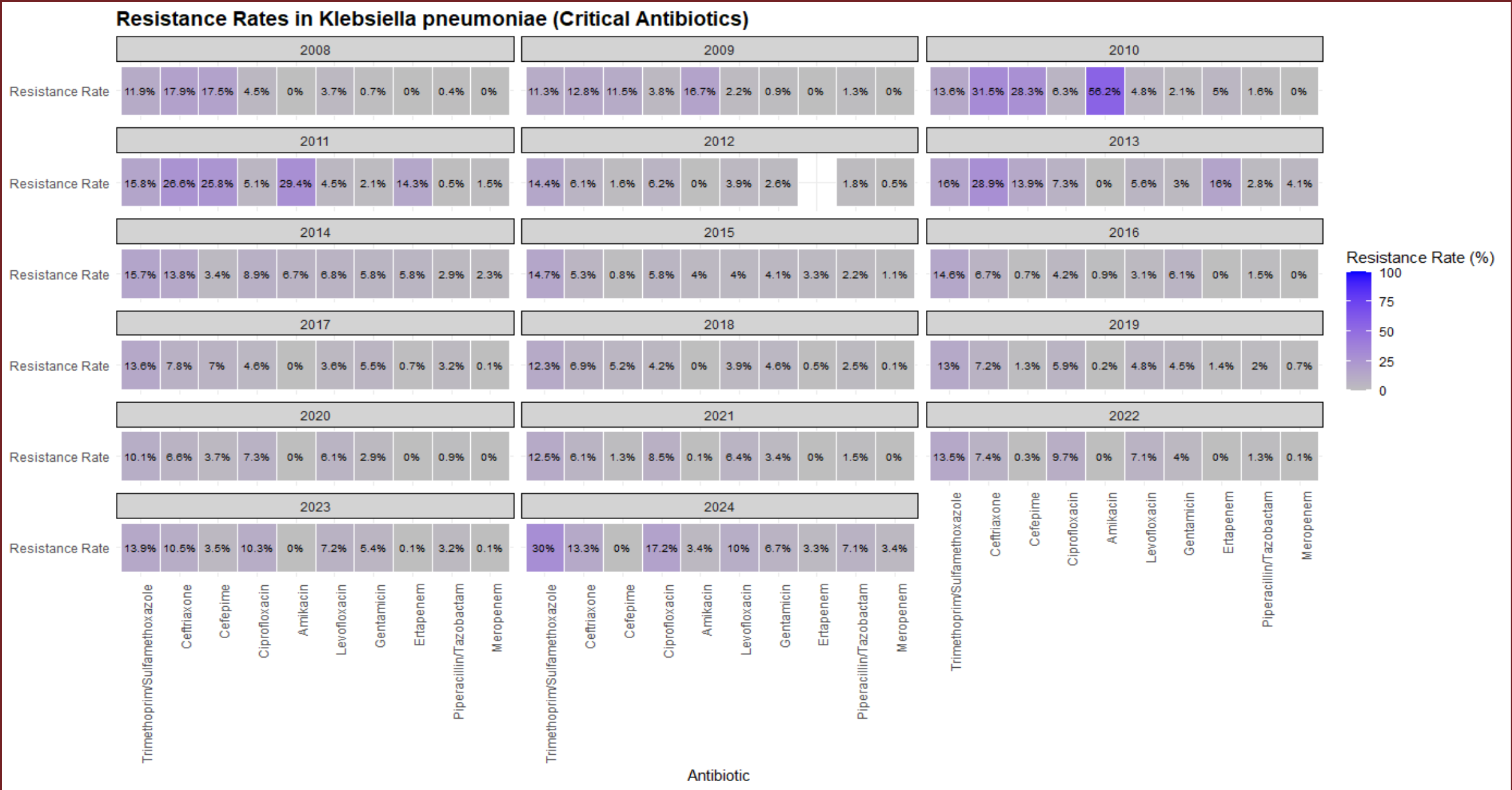


M.A. Kosinski

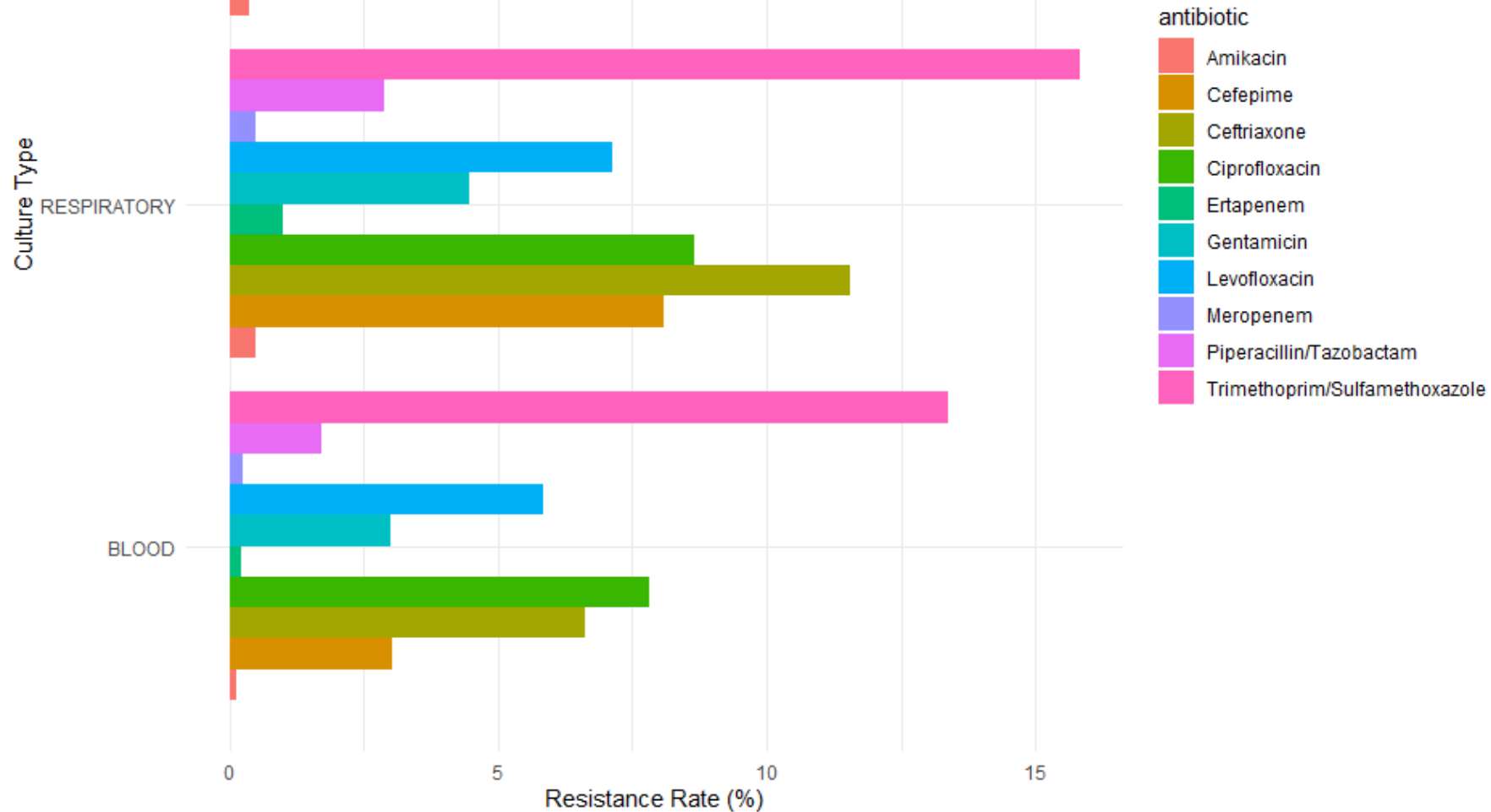
Resistance Trends for Critical Antibiotics in *K. pneumoniae*



TEMPORAL TRENDS IN KLEBSIELLA PNEUMONIAE RESISTANCE RATE



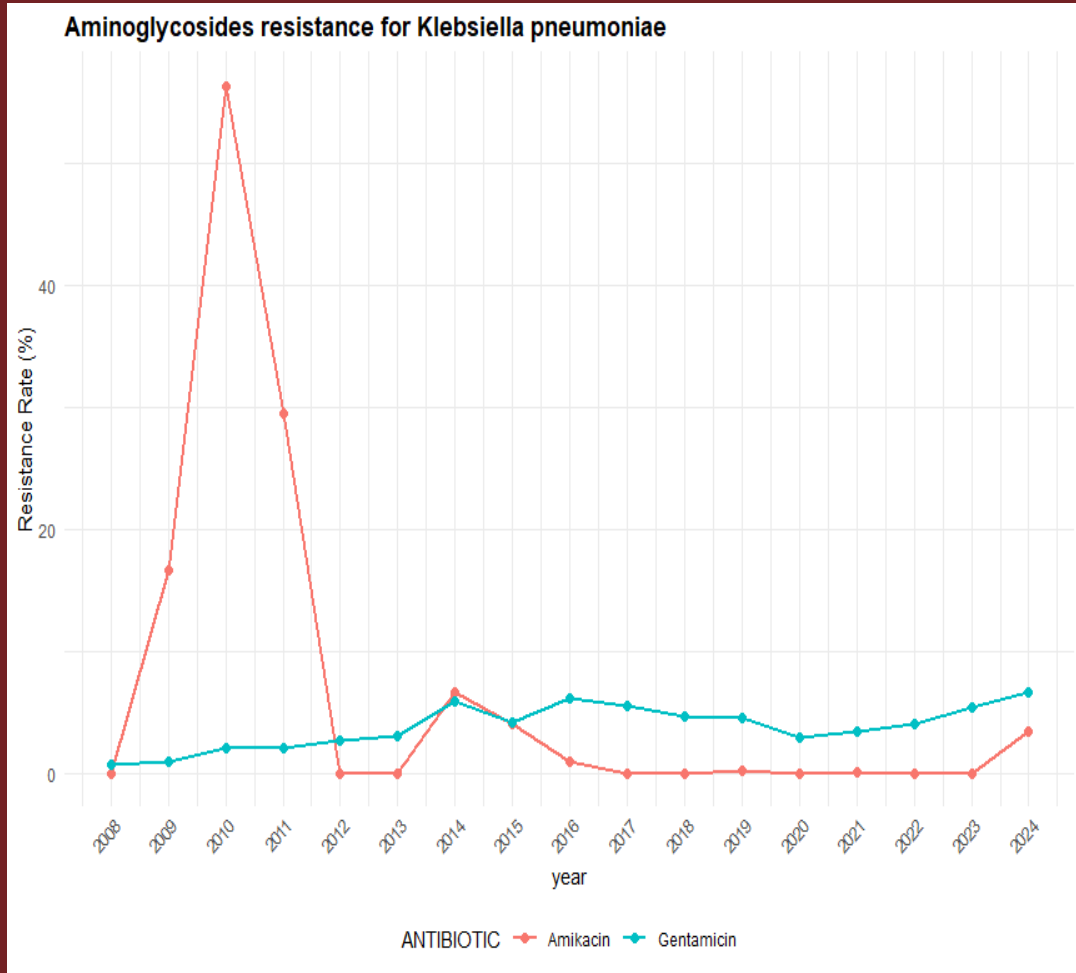
Resistance Rates by Culture Type



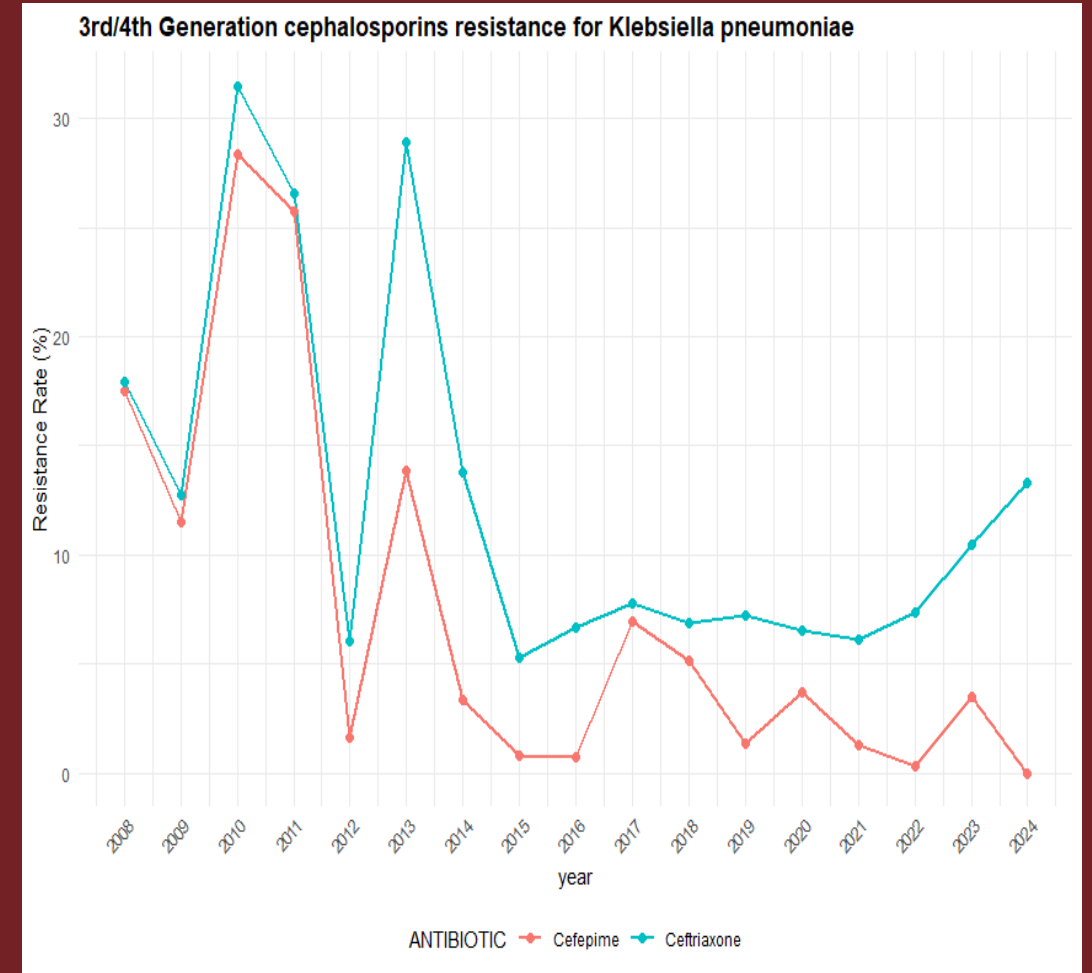
DISTRIBUTION
OF RESISTANCE
RATE FOR
KLEBSIELLA
PNEUMONIAE
ACCORDING TO
CULTURE TYPE

ANTIBIOTIC RESISTANCE ACCORDING TO CLASSES

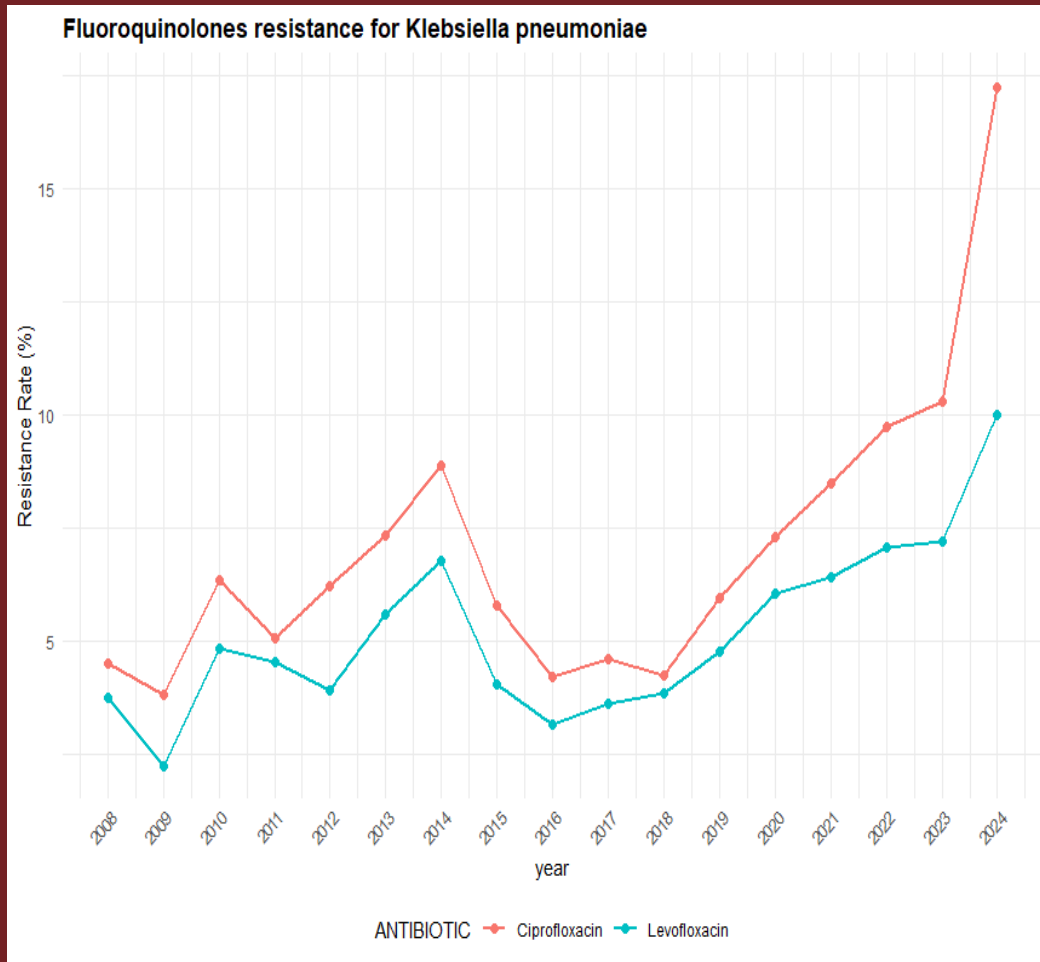
1. AMINOGLYCOSIDES



2. 3RD/4TH GENERATION CEPHAOSPORINES



3. FLUOROQUINOLONES



4. FOLATE PATHWAY INHIBITORS

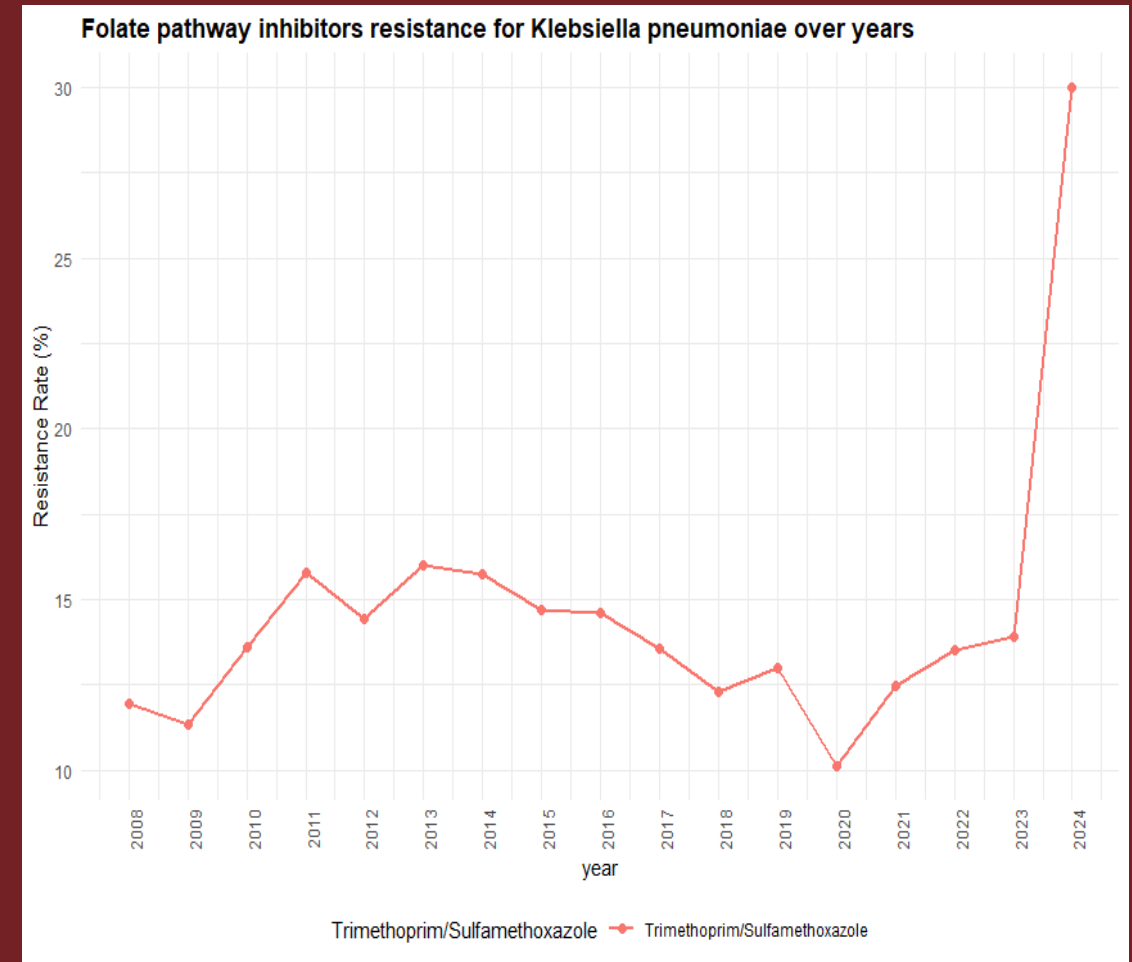
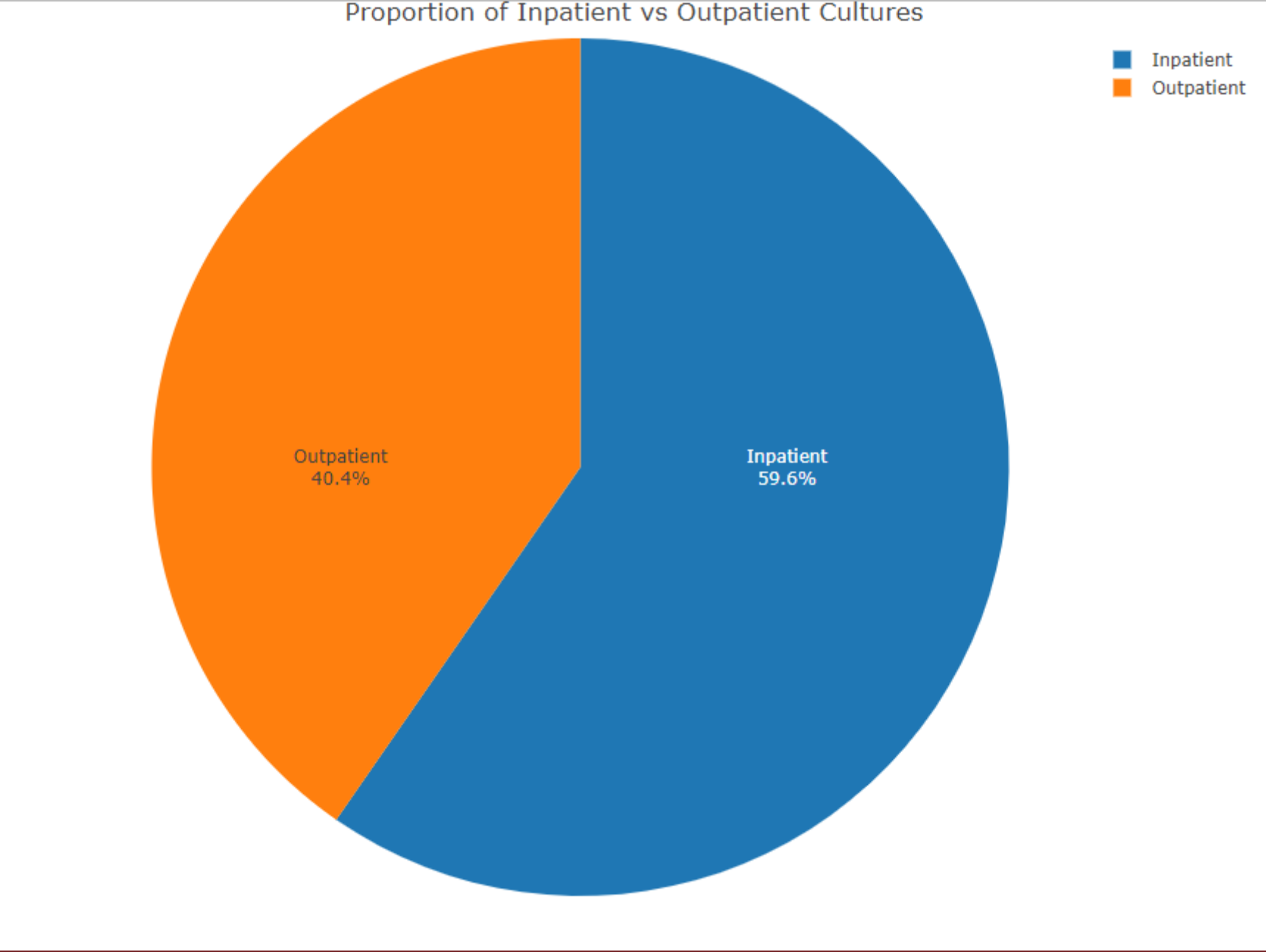
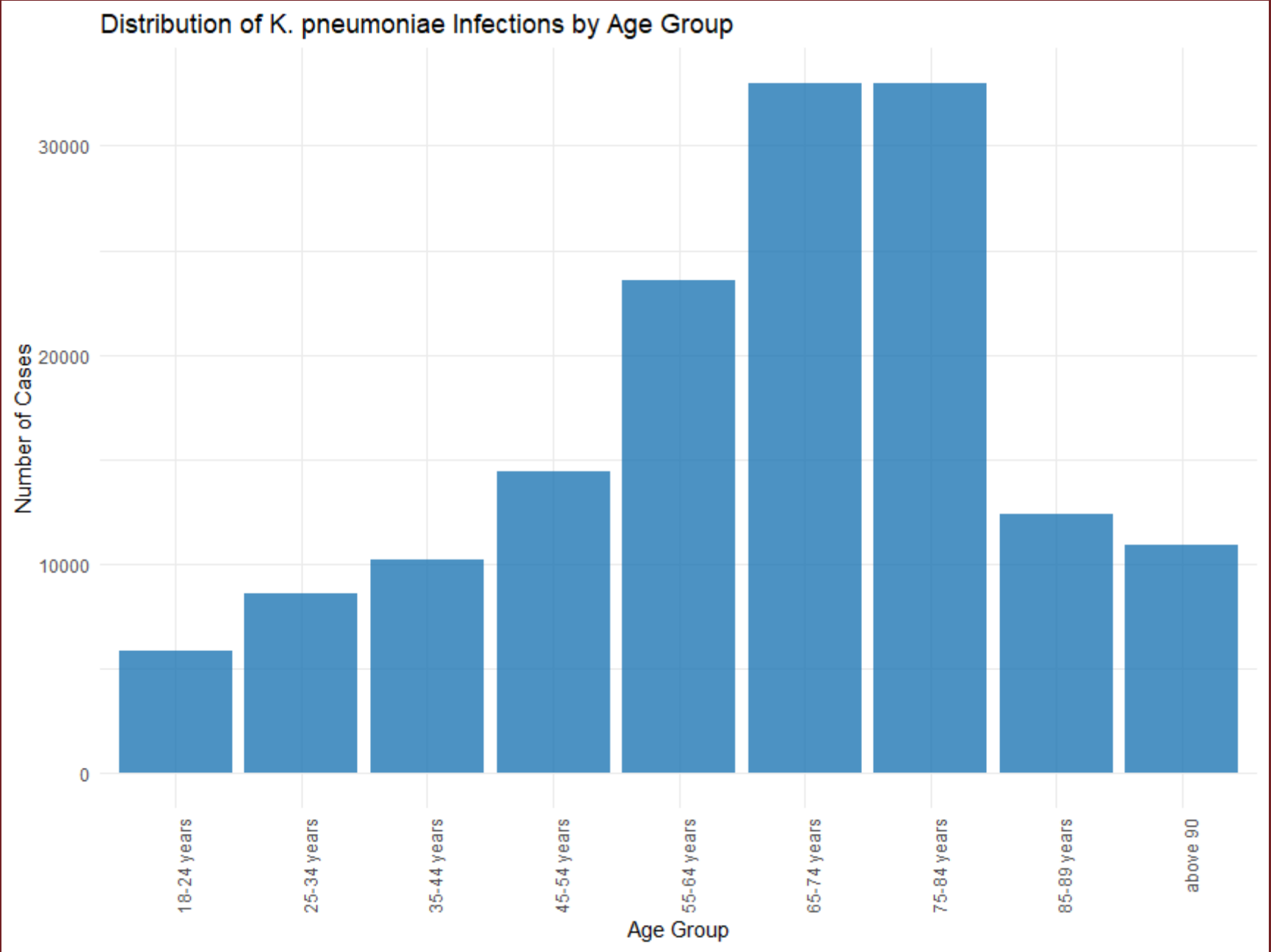


ILLUSTRATION OF THE
PROPORTION OF
INPATIENT VS
OUTPATIENT CULTURES
AND THE CORRELATION
BETWEEN
K.PNEUMONIAE AND
NOSOCOMIAL
INFECTION



VISUALIZING THE
RELATIONSHIP
BETWEEN AGE AND
KLEBSIELLA
PNEUMONIAE
INFECTION

These observations may reflect differences in the selected populations studied. It is not surprising that advanced age was associated with an increased risk because this has been a fairly consistent observation across many previous studies and may be reflective of an age-associated increased risk in general for the comorbid illnesses highly associated with increased risk.



KEY MESSAGE

1. This project showed that the overall antibiotic resistance rate of *Klebsiella pneumoniae* increased significantly compared to 10 years previously.
2. - Broad-spectrum antibiotics are more commonly used for empirical treatment of both community-acquired and healthcare-associated infections compared to 10 years ago.
3. - Unnecessary use of broad-spectrum antibiotics can be reduced if *K. pneumoniae* bacteremia is suspicious and there are no healthcare risk factors, because antibiotic resistance in community-acquired *K. pneumoniae* bacteremia was not increased.

REFERENCES

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5. Shiri Navon-Venezia, Kira Kondratyeva, Alessandra Carattoli, *Klebsiella pneumoniae: a major worldwide source and shuttle for antibiotic resistance, FEMS Microbiology Reviews*, Volume 41, Issue 3, May 2017, Pages 252–275, <https://doi.org/10.1093/femsre/fux013>

THANK YOU



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<https://github.com/meghalawy/R-EGcompBio-/blob/main/K.P%20project>